



MINING DEALER BEST PRACTICE

# Elimination of Live Work - Checking HMS Slew Ring Tilting Play

Maintenance and  
Repair

Application

Component Life  
Management

Component  
Renewal (CRC)

MARC  
Management

|                                                   |    |
|---------------------------------------------------|----|
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May 2023  
1.11-230525-1

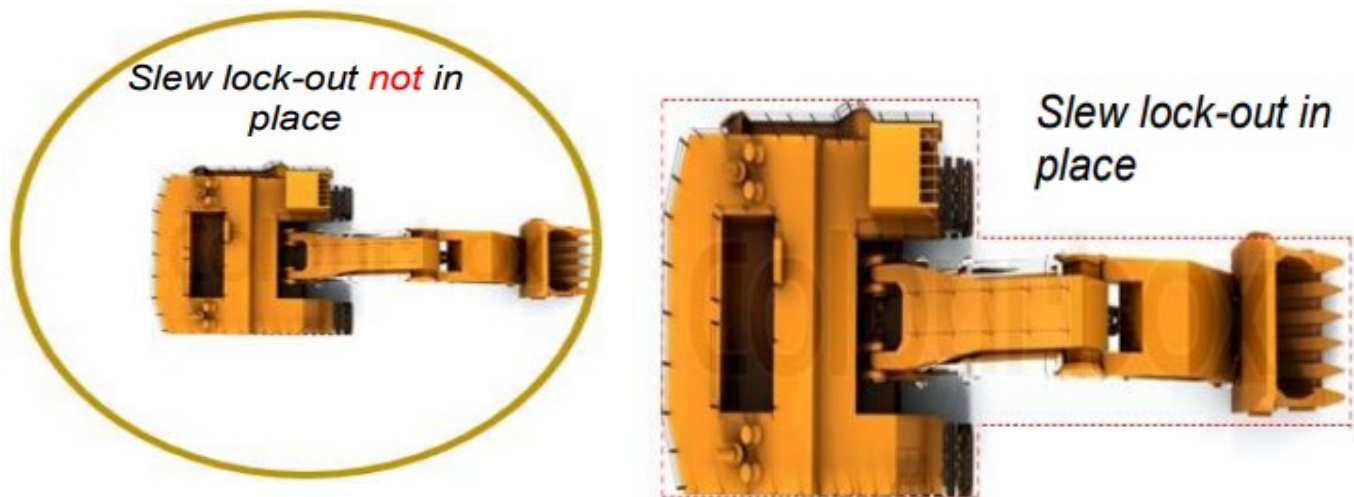
## 1.0 Introduction

**Live Work:** Any task (i.e. testing, commissioning or troubleshooting) that requires the system to be energized and the person to enter the footprint of the machine (excludes doing the same task from the machine cabin or outside the machine footprint).

This “Elimination of Live Work (ELW) Procedure” replaces the corresponding description in the HMS machine documentation. The intention of the procedure is to eliminate live work by removing the technician from the area of the machine to reduce possible hazard of injury while inspecting or maintaining the machine.

This procedure provides the information to measure the tilting play of the swing gear. The tilting play must be measured at machine commissioning and used as a reference to measure increase of play.

This procedure can be done using Cat Multitool GP to eliminate the risk of proximity to the machine.



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## 2.0 Best Practice Description

This applies when checking the HMS slew ring tilting play for all machine listed below.

### Elimination of Live Work

#### Task: Check Slew ring Tilting Play

| Model Type | Prefix          | Serial Number | Model Type | Prefix          | Serial Number |
|------------|-----------------|---------------|------------|-----------------|---------------|
| 6015       | DHB             | all           | 6015 FS    | DHF             | all           |
| 6018       | DHT             | all           | 6018 FS    | DHD             | all           |
| 6030       | DHL, J7F, J8F   | all           | 6030 FS    | DHG, J7F, J8F   | all           |
| 6040       | DHJ<br>CC9, ELZ | all           | 6040 FS    | DHX<br>CC9, ELZ | all           |
| 6050       | DHM             | all           | 6050 FS    | DH2             | all           |
| 6060       | DHN             | 60001-60190   | 6060 FS    | DH3             | 60001-60194   |
| —          | —               | —             | 6090 FS    | DHS, DM6        | all           |

### Important Safety Information

Operators and maintenance personnel should be protected in the best possible way while working on the machine. Therefore, all work should only be carried out (if not described differently):

- ☐ not with and on running engines / motor
- ☐ not while the machine is moving
- ☐ not with moving work equipment
- ☐ not under raised work equipment
- ☐ not on systems under pressure
- ☐ not on components that are too hot to be touched by hand with gloves.

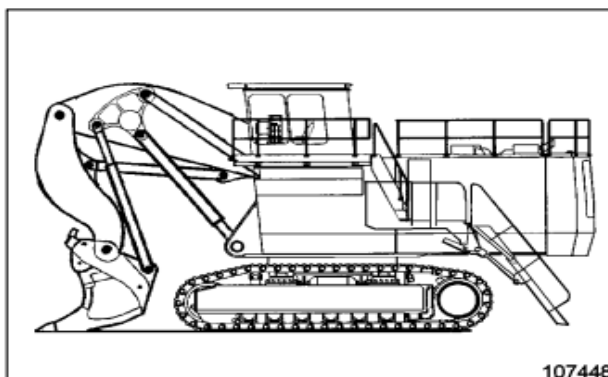


Fig. 1: Servicing position for FS machines

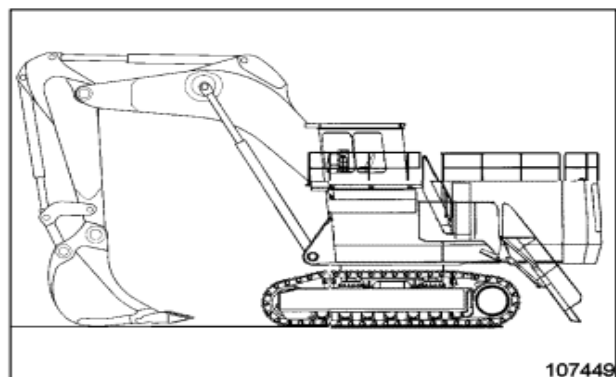


Fig. 2: Servicing position for BH machines

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### 3.0 Implementation Steps

#### Tools Required:

| Required Parts |                         |                                  |
|----------------|-------------------------|----------------------------------|
| Qty            | Part Number             | Description                      |
| 8              | 576-0855                | Measurement Tool Gp              |
| 8              | 285-0898 <sup>(1)</sup> | Electronic Position Indicator Gp |
| 1              | 515-8465 <sup>(2)</sup> | Multi-Tool Gp                    |
| 1              | 6V-6042 <sup>(3)</sup>  | Indicator Point Gp               |

#### Job Preparation:

1. Isolate the machine before positioning the Tool GP on it.
2. Allow only authorized people to perform the job procedures.
3. Job needs 3 people to perform it (Operator, Spotter, and Inspector).
4. Ensure that all slew ring bolts are tightened as described in the Service Manual.

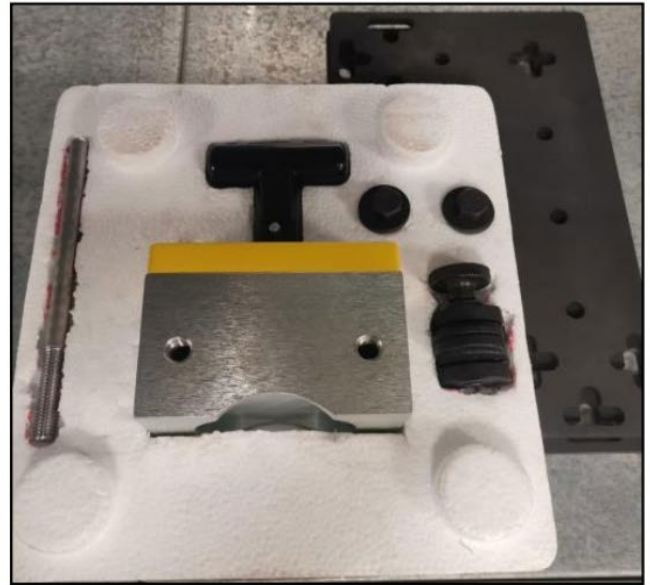


#### Job Procedure:

1. Assemble Tool GP (576-0855).
2. Assemble magnet tool (1) to Stud (3).
3. Assemble stud (3) to swivel post snug (6).
4. Assemble sensor to swivel post snug (6), after magnet tool (1) is located into position on the slew ring.

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| Kit Consist 576-0855 Measurement Tool Gp |     |             |                  |
|------------------------------------------|-----|-------------|------------------|
| Item                                     | Qty | Part Number | Description      |
| 1                                        | 1   | 570-4825    | Magnet Tool      |
| 2                                        | 1   | 570-4826    | Bracket          |
| 3                                        | 1   | 570-4827    | Stud             |
| 4                                        | 2   | 0S-0509     | Bolt             |
| 5                                        | 4   | 9L-9132     | Washer           |
| 6                                        | 1   | 7H-1948     | Swivel Post Snug |
| 7                                        | 2   | 9S-8752     | Hex Nut          |



5. Attach measurement tool group to the Slew Ring by holding the magnet tool against the slew ring component then turn the handle counterclockwise a full 180 degrees.

**Note:** Turning the magnet tool handle counterclockwise 180 degrees activates the magnet and turning the magnet tool handle clockwise 180 degrees de-activates the magnet.

**Note:** Ensure that the areas where the measurement tool group attaches to the machine are clean and free of debris.



**Note:** Ensure that the magnet tool has full contact against the component.

6. Adjust the locations of the measurement tool group, adjust the location of stud using the slotted hole on the bracket, or change the sensor tip using the 6V-6042 Indicator Point GP to ensure that the sensor tip is against the component.

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**Note:** Ensure that casting imperfections or locators do not interfere with the sensor tip. If necessary, adjust the location of the sensor, failure to adjust the location may invalidate the results of the test.

**Note:** Ensure that magnet tool has full contact against the component to receive an accurate measurement, the sensor should be perpendicular to Under Carriage.

7. Attach the extension cable to the sensor.

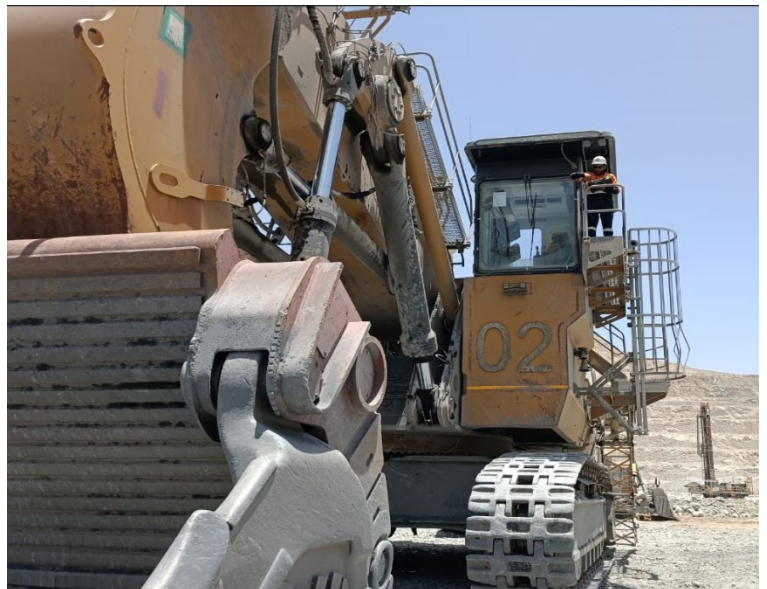
8. Route the extension cable to a location where the technician can perform the inspection inside the cab to be away from the line of fire.

9. Secure the extension cable using tie straps or other means so that the connections do not come loose, and the cable are protected during the test.

10. Connect the extension cable lead to the Cat Multi-Tool.



Cable Extensions



Inspector arrangement

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Inspector performing the inspection inside the cab



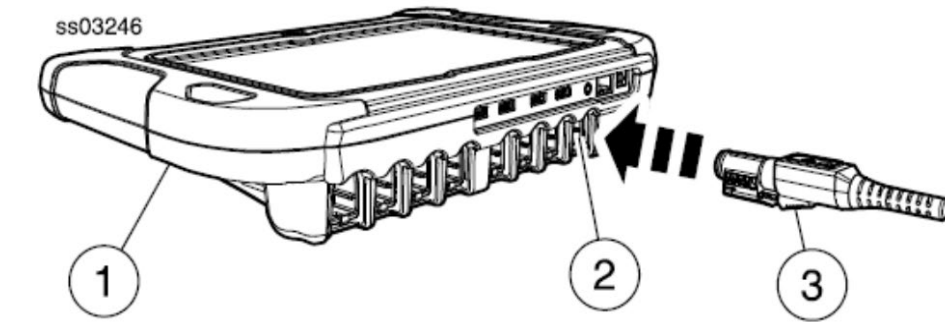
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Elimination of Live Work - Checking HMS Slew Ring  
Tilting Play

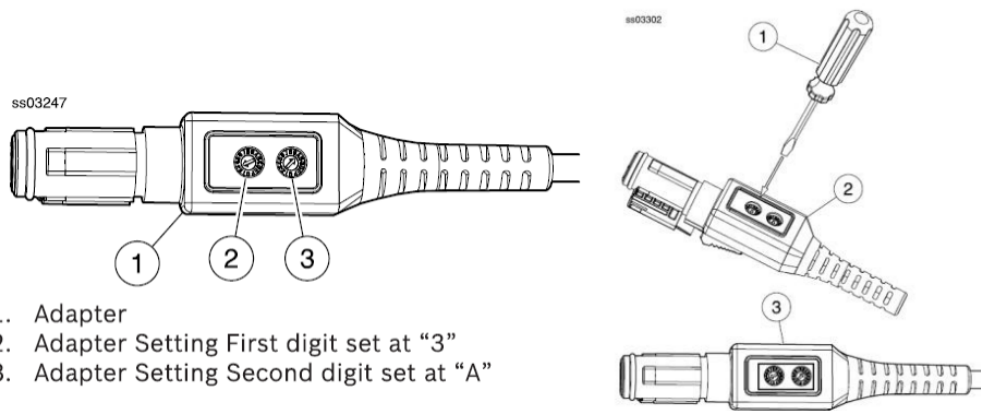
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1. Multi Tool Plus
2. Sensor Port



1. Adapter
2. Adapter Setting First digit set at "3"
3. Adapter Setting Second digit set at "A"

*Note: The adapter is set to 3,A*

1. Small screwdriver
2. Adapter
3. Adapter is set to 3, A

11. Adjust the adapter to the correct position.

|         |     |       |                 |
|---------|-----|-------|-----------------|
| 8T-1002 | 3,8 | Black | Position Sensor |
|---------|-----|-------|-----------------|



12. Turn ON the Cat Multi-Tool and configure for the electronic position sensor test.

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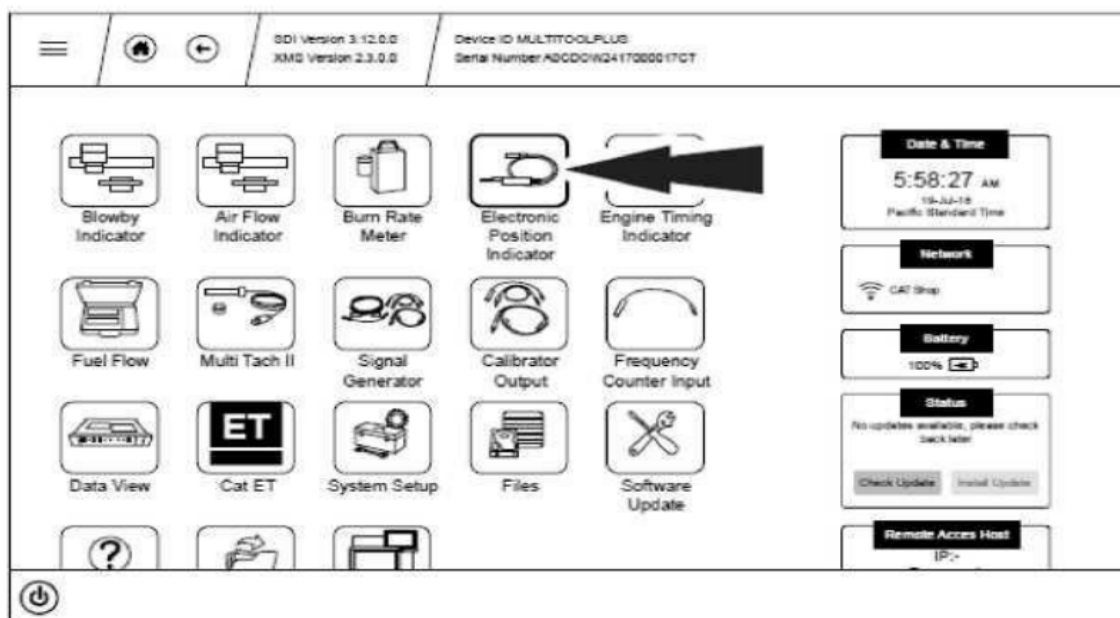


13. Verify that all personnel are accounted for and prepare the machine for maintenance. Refer to the appropriate machine-specific Operation and Maintenance Manual, Prepare the Machine for Maintenance.

“The Technician conducting the inspection is out of the live zone of the machine”

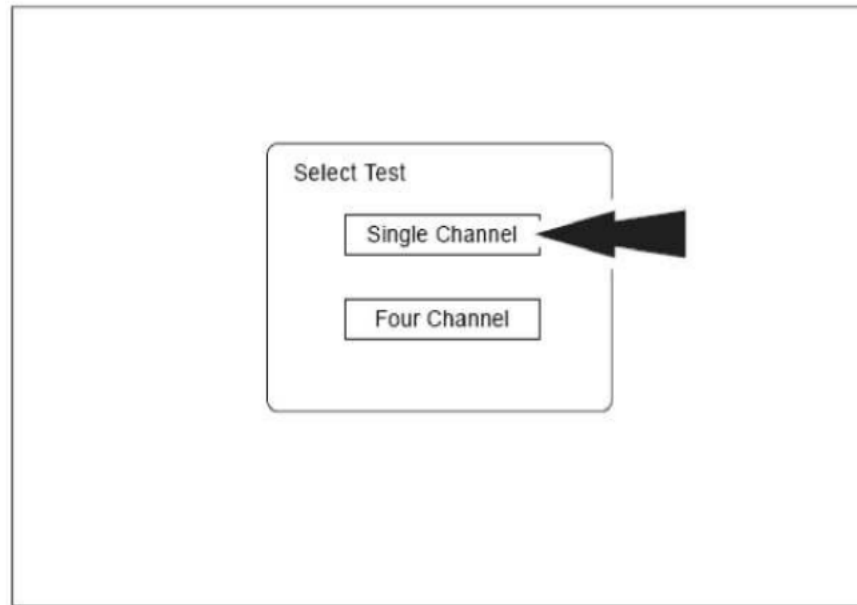


## Electronic Position Indicator Test and Cat Multi-Tool Screen Shots

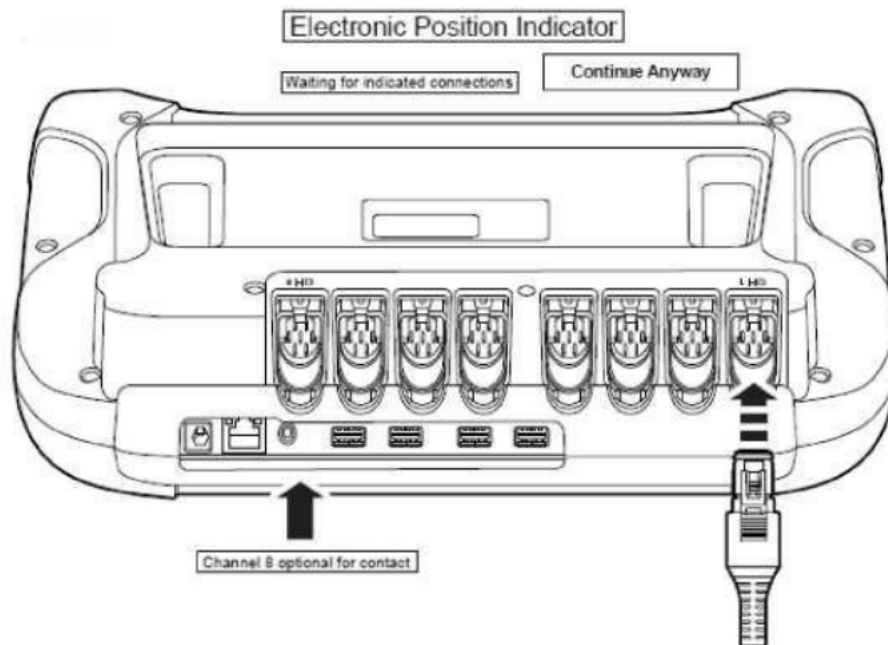


- Select, "Electronic Position Indicator Test" from the home screen.

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- Select single or four channels.



- Verify that the correct sensors and adapters are being used.

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- Verify that the adapter has the correct ID set. Refer to the matrix in the Appendix in Tool Operating Manual, NEHS1278, "515-8465 Multi-Tool Plus".

**Note:** Once the Multi Tool Plus has the adapter and correct Cable ID set, the Multi Tool will automatically go to the next screen. However, the user can select "Continue Anyway" to move to the next screen.

- Set the position sensor to zero in the Cat Multi-Tool.



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- From More menu select (Zero).



## Measure the Slew ring Tilting Play.

14. Place the machine on level ground
15. Empty the bucket
16. Position the superstructure parallel with the tracks
17. Position the machine as shown in fig. 5

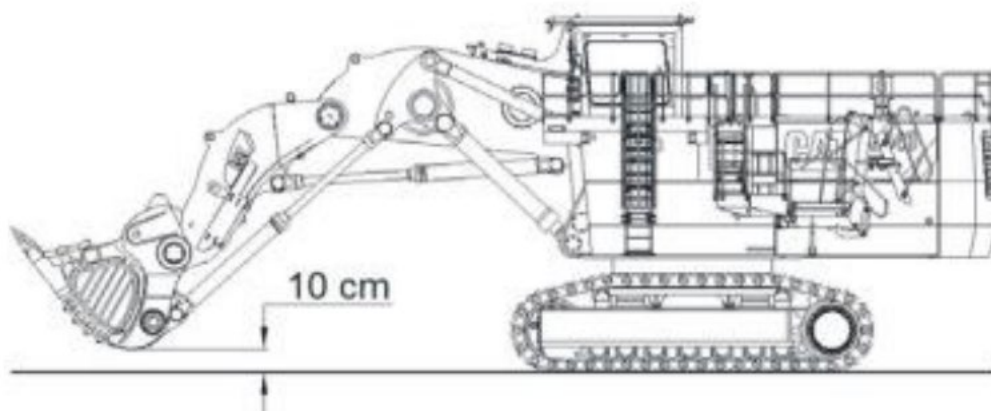


Fig.5: position the machine attachment as shown. Ensure, that there is a distance of about 10 cm between the attachment and the ground. Zero gauge.

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18. Position the Sensor and GP. on the slew ring in location 1 (fig. 4)

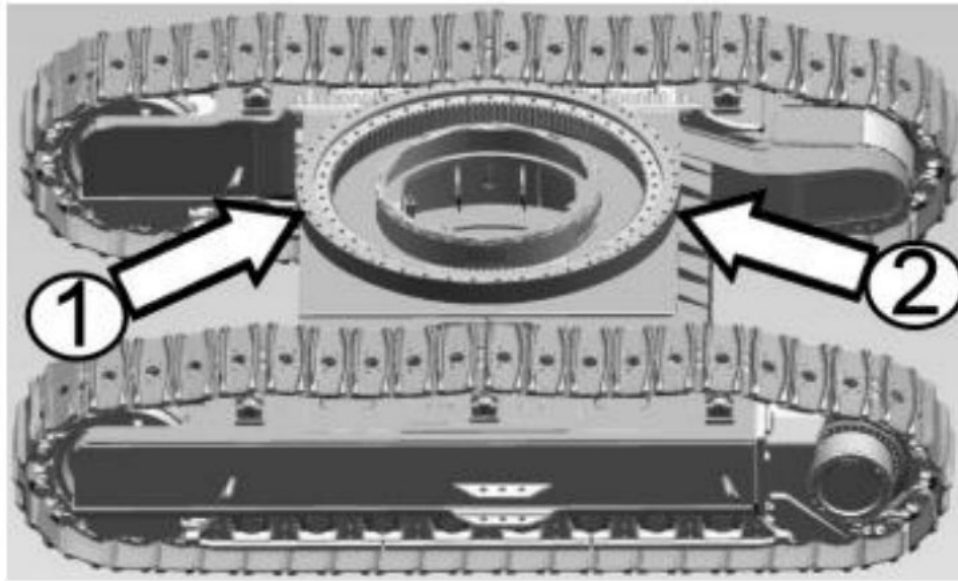


Fig.4: top view of location of the dial gauge  
location 1 = centered on idler side  
location 2 = centered on sprocket side



19. Set Multi-Tool Plus gauge to zero.  
20. Position the machine as shown in fig. 6.

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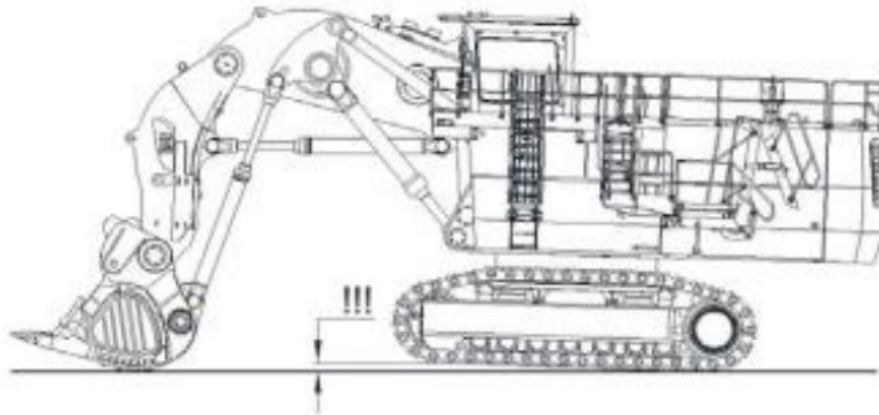


Fig.6: position the machine as shown. Ensure, that the tracks are slightly lifted off the ground. Take measurement.

21. Read gauge and determine the amount of swing ring tilting play. Record value in form.

|                                                   |  |                                          |                                                       |                                            |                                                       |
|---------------------------------------------------|--|------------------------------------------|-------------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| <b>CATERPILLAR®</b>                               |  | Check Tilting Play of Slew Ring - Form   |                                                       | Caterpillar Global Mining HMS GmbH         |                                                       |
|                                                   |  |                                          |                                                       | Media Number EM030509-02                   |                                                       |
| <b>Position of dial gauge</b><br>                 |  | Machine-Typ<br>Serial-No                 | Op.-Hrs.                                              | Date                                       | PN slew ring                                          |
|                                                   |  |                                          |                                                       |                                            |                                                       |
|                                                   |  | serial no. slew ring                     |                                                       |                                            |                                                       |
|                                                   |  | First Measurement<br>Superstructure 0°   |                                                       | Second Measurement<br>Superstructure 180 ° |                                                       |
|                                                   |  | 1st.<br>measure<br>gauge on<br>boom side | 2nd.<br>measure<br>gauge on<br>counter weight<br>side | 3rd.<br>measure<br>gauge on<br>boom side   | 4th.<br>measure<br>gauge on<br>counter weight<br>side |
| <p>Distance bucket to ground = 10 cm</p>          |  | Position of<br>measure*                  | adjust 0 on<br>gauge                                  |                                            |                                                       |
| <p>Tracks are slightly lifted off the ground.</p> |  | measure<br>play in mm                    |                                                       |                                            |                                                       |

22. Position the Sensor and GP. on the slew ring in location 2 (fig.4).

|                                                                                                                 |  |                         |                 |
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23. Repeat steps 1 to 22.

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Caterpillar Global Mining HMS GmbH

Media Number EM030509-02

Tilting Play of Slew Ring - Datasheet

| Machine Type       | Part-No.   |            |               | Tilting Play of new Slew-Ring (measured on site during commissioning) in mm* | Max. increase of Tilting Play in mm** |
|--------------------|------------|------------|---------------|------------------------------------------------------------------------------|---------------------------------------|
|                    | (Navision) | (SAP)      | (Caterpillar) |                                                                              |                                       |
| Cat 6015 / 6015 FS | 2 270 612  | BU_4020858 | 541-5432      |                                                                              | 1,00                                  |
| Cat 6018 / 6018 FS | 1 975 679  | BU_4019561 | 535-3219      |                                                                              | 1,20                                  |
| Cat 6030 / 6030 FS | 2 450 905  | BU_1457293 | 471-0660      |                                                                              | 1,70                                  |
| Cat 6040 / 6040 FS | 2 450 745  | BU_1588211 | 488-5914      |                                                                              | 2,00                                  |
| Cat 6040 / 6040 FS | 3 683 374  | BU_4592047 | 568-5272      |                                                                              | 2,00                                  |
| Cat 6050 / 6050 FS | 2 450 745  | BU_1588211 | 488-5914      |                                                                              | 2,00                                  |
| Cat 6060 / 6060 FS | 2 450 745  | BU_1588211 | 488-5914      |                                                                              | 2,00                                  |
| Cat 6090 FS        | 2 763 220  | BU_1522910 | 534-1824      |                                                                              | 2,50                                  |

\* during commissioning, take first measurements as described in instruction. This value is used as base reference for the play. \*\* max. permissible tilting play = tilting play new (to be measured on new machine) + max. increase of tilting play

\*\* example Cat 6030: play new = 0,16 mm, increase of play = 1,7 mm // max. permissible play = 1,86 mm

Attention: slew ring bolts and washers are for single-use only.

#### 4.0 Benefits

- ☐ It improves the safety of the task.
- ☐ incremental business for Cat technology.
- ☐ transfer Cat technology to the customers.

#### 5.0 Resources Required

| Required Parts |                         |                                  |
|----------------|-------------------------|----------------------------------|
| Qty            | Part Number             | Description                      |
| 8              | 576-0855                | Measurement Tool Gp              |
| 8              | 285-0898 <sup>(1)</sup> | Electronic Position Indicator Gp |
| 1              | 515-8465 <sup>(2)</sup> | Multi-Tool Gp                    |
| 1              | 6V-6042 <sup>(3)</sup>  | Indicator Point Gp               |

#### 6.0 Supporting Attachments / References

Swing Bearing Movement - Measure (i08787866)

#### 7.0 Related Best Practices

NIA

|                                                                                                                 |                   |              |                         |
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## 8.0 Acknowledgements

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